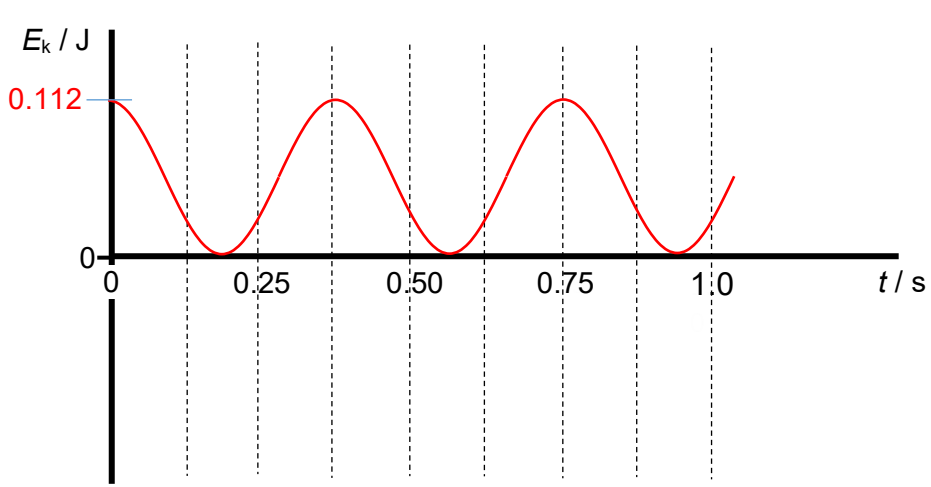
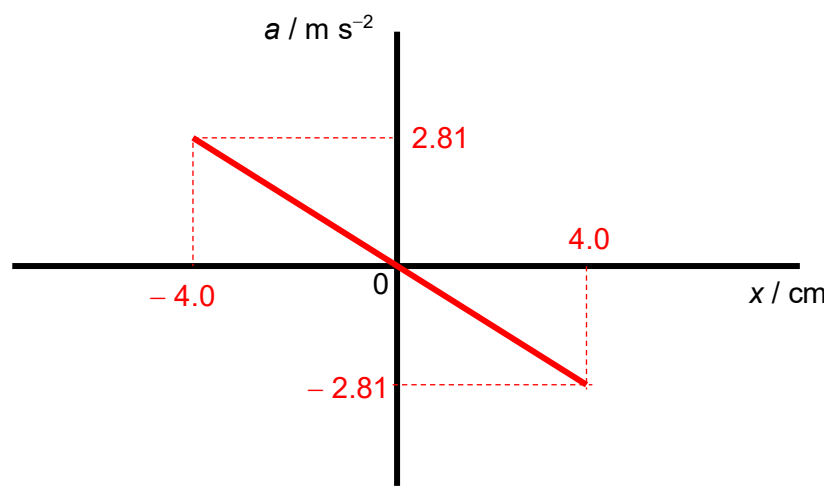


			No mark will be awarded for (0) as it is given in the question in (ii)	[5]
4	(a)		$T = 0.75 \text{ s}$ $\omega = 2\pi / T$	

		$= 8.38 \text{ rad s}^{-1}$	1
(b)		$a_o = \omega^2 x_o$ $= 8.38^2 (4.0 \times 10^{-2})$ $= 2.81 \text{ m s}^{-2}$	1
(c)		At the lowest point of oscillation $\Sigma F = ma_o$ $T - mg = ma_o$ $25 - m(9.81) = m(2.81)$ $m = 1.98 \text{ kg}$	1 1
(d)		 Max $E_K = \frac{1}{2} m (x_o \omega)^2$ $= \frac{1}{2} (2.0)(0.04)^2(8.38)^2$ $= 0.112 \text{ J}$ Labelling of the correct maximum E_K [1] Correct positions of maximum E_K and shape of the graph [1]	2
(e)		 Correct straight line with negative gradient passing through origin [1] Correct labelling of the critical values and no exceeding of line beyond 4.0 cm [1]	2